SIMATS ENGINEERING

ASSESSMENT TOOL 2:INDUSTRYPROBLEMSOLVING TASK

Cloud ComputingandBigDataAnalytics–CSA1518

CO4: Analyze big data characteristics and challenges across domains including security and application aspects.

Question Count: 5 | Total Marks: 40 | Duration: 60 Minutes

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Question 1: Retail Data Optimization Problem

A retail chain operates physical stores and an online platform and generates large volumes of sales, inventory, and customer-behavior data. The required solution must support real-time inventory tracking, demand forecasting, and personalized marketing recommendations.

1. Industry Context and Requirements

Integrate sales, inventory, online activity, and customer behavior data.
Process incoming data with low latency for real-time inventory visibility.
Use historical data for demand forecasting.
Analyze customer behavior for personalized marketing.
Provide scalable storage and distributed processing for increasing data volumes.

2. Proposed Big Data Architecture

Retail Big Data Architecture

POS / Online Platform	Data Ingestion	Stream + Batch Processing	DataLake / Warehouse	Analytics & ML	Inventory / Marketing Dashboard
↓	↓	↓	↓	↓	

The architecture combines streaming and batch processing. Store POS systems and the online platform produce sales, inventory, and customer-event data. A distributed ingestion layer collects the data. Stream processing provides current inventory information, while batch processing analyzes historical data.

3. Tools, Technologies and Models

Component	Suitable technology / approach	Purpose
Data ingestion Stream	Apache Kafka	Collect high-volume real-time events
processing Batch	Apache Spark Structured Streaming	Process inventory and customer events
processing	Apache Spark	Process historical sales and behavior data Store raw and processed datasets
Storage	Distributed data lake / cloud object storage	Analyze sales and inventory patterns Predict
Analytics	Spark SQL / Big Data analytics Time-	future product demand Generate targeted
Demand forecasting	series forecasting models	marketing recommendations
Personalization	Recommendation / customer-segmentation models	

4. Working

1. POS and online systems generate sales, inventory, and customer-behavior events.
2. The ingestion layer collects the events continuously.
3. Stream processing updates inventory information with low latency.
4. Historical data is stored for large-scale batch analysis.
5. Forecasting models estimate future demand.
6. Customer analytics identifies preferences and behavior patterns.
7. Recommendation models generate personalized offers.
8. Dashboards provide inventory and marketing insights to business users.

5. Security and Risks

Protect customer information using access control and encryption.
Apply authentication and authorization to data and analytics services.
Maintain data-quality checks to avoid incorrect inventory forecasts.
Use backups and monitoring to reduce operational risks.

6. Expected Outcome

Thesolution providesreal-time inventory visibility, supports data-driven demand forecasting, and enables personalized marketing while remaining scalable as the retail data volume grows.

Question 2:SocialMediaFraudDetectionProblem

A social media platform processes billions of posts, comments, likes, and shares every day and needs to detect fake accounts and misinformation quickly. Therefore, the solution must support scalable stream processing and machine-learning-based detection.

1. Industry Context and Requirements

Process very high volumes of user interactions.
Detect suspicious account behavior with low latency.
Identify potentially misleading or false information.
Support continuous machine-learning analysis.
Scale processing as platform activity increases.

2. Proposed Big Data Architecture

SocialMedia Fraud Detection Architecture

User Interactions ↓	Kafka / Stream Ingestion	Stream Processing	Feature Extraction	MLDetection Models	Alerts / Moderation
	↓	↓	↓	↓	

3. Technologies and ML Approach

Component	Suitable technology / approach	Purpose
Event ingestion	Apache Kafka	Capture posts, likes, shares and other events
Stream processing Storage	Apache Spark Structured Streaming Distributed data lake	Process interaction streams continuously
Feature engineering	Distributed feature processing	Retain historical interactions and features
Fake-account detection	Classification / anomaly detection	Generate behavioral indicators
Misinformation detection	Text/content classification models	Identify suspicious account patterns
Monitoring	Real-time dashboards and alerts	Flag potentially misleading content
		Support rapid moderation decisions

4. Working

9. User interactions are continuously collected as events.
10.Stream processing analyzes events without waiting for a large batch.
11.Behavioral features such as unusual activity patterns are extracted.
12.Machine-learning models calculate fraud or misinformation risk. 13.High-risk accounts or content are sent to moderation/alert systems.
14.Historical datais used to retrain and improve the models.

5. Security and Risks

Protect user data and restrict access to sensitive information.
Monitor false positives and false negatives in automated detection. Prevent unauthorized manipulation of detection models or training data. Maintain audit logs for moderation decisions.

6. Expected Outcome

Astreaming BigData architecture enables rapid detection of suspicious behavior while distributed processing and machine learning allow the system to scale to very high interaction volumes.

Question 3:GovernmentDataPlatformProblem

A government agency is building a national data platform combining census, economic, and geospatial datasets. Different departments need shared, integrated access while strict privacy and security requirements mustbe maintained.

1. Industry Context and Requirements

Integrate datasets from multiple departments and domains.
Provide interoperability between agencies.
Maintain common data definitions and standards.
Control access to sensitive government information.
Ensure data quality,privacy, auditability, andaccountability.

2. Proposed Governance and Data Management Architecture

Government Data Platform Architecture

Department Data Sources	Secure Ingestion	Integrated Data Lake	Metadata + Governance	Analytics / Shared APIs	Authorized Departments
↓	↓	↓	↓	↓	

3. Governance Strategy

Area	Strategy
Data standards	Use common schemas, formats, identifiers, and metadata standards
Data ownership	Assign clear data owners and custodians for each dataset Use
Access control	role-based access and least-privilege permissions Apply
Data quality	validation, deduplication, consistency and completeness checks Maintain a searchable catalog describing datasets and lineage
Metadata Privacy	Apply masking, anonymization/pseudonymization where appropriate

Security	Use encryption, authentication, monitoring and audit logs Use
Interoperability	standardized APIs and common data exchange formats

4. Working

15.Departments provide census, economic, and geospatial datasets.
16.Secure ingestion validates and transfers the data.
17.Data is integrated into a shared platform using common standards.
18.Metadata records describe sources, ownership, quality, and lineage. 19.Governance policies control who can access each dataset.
20.Authorized departments access integrated data through controlled services or APIs.
21.Auditingand monitoring track important access and data-management activities.

5. Risks and Controls

Unauthorized access → strong authentication, authorization, and auditing. Privacy leakage → data minimization and appropriate masking/anonymization. Inconsistent datasets → common standards and quality validation.
Data misuse → clear ownership, policies, access reviews, and audit trails.

6. Expected Outcome

Thestrategycreatesa governed and interoperable national data platform where departments can share useful information while maintaining privacy, security, quality, and accountability.

Question 4: Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute and requires real-time fraud detection using bothhistoricaland streaming transactiondata. Low latency and high accuracy are critical.

1. Industry Context and Requirements

Process millions of transactions with very low latency.
Combine real-time transaction streams with historical customer behavior. Detect suspicious transactions before or immediately after authorization. Maintain high detection accuracy.
Scale processing during high transaction periods.

2. Proposed Big Data Pipeline

Banking Fraud Detection Pipeline

Transaction Stream	**Kafka / Ingestion**	**Stream Processing**	**FeatureStore+ Historical Data**	**Fraud ML Model**	**Risk Score / Alert**
↓	↓	↓	↓	↓	

3. Technologiesand Algorithms

Component	**Suitable technology / approach**	**Purpose**
Streaming ingestion	Apache Kafka	Receive transaction events continuously Compute real-time transaction features
Stream processing	Apache Spark Structured Streaming	Store previous transactions and customer
Historical storage	Distributed data lake / warehouse	history Calculate behavioral and transaction
Feature engineering	Distributed processing	indicators Identify suspicious transaction patterns
Fraud algorithm	Classification and anomaly detection	

Decision layer	Risk scoring / rules	Combine model score with business rules
Alerting	Real-time alert service	Notify fraud systems or investigators

4. Working

22.Atr ansaction is received by the streaming ingestion layer.
23.The transaction is processed immediately.
24.Real-time features are combined with relevant historical customer behavior. 25.The fraud model calculates a risk score.
26.Business rules and risk thresholds are applied.
27.High-risk transactions generate alerts or additional verification.
28.Historical outcomes are retained for model evaluation and improvement.

5. Security and Risks

Encrypt sensitive transaction data in transit and at rest.
Use strict access controls for financial data.
Maintain audit logs for fraud decisions and investigator actions.
Monitor model performance to reduce false positives and false negatives.
Design for fault tolerance so fraud detection continues during infrastructure failures.

6. Expected Outcome

Thepipelinecombinesreal-time streaming with historical analytics to provide fast and accurate fraud detection while supporting large transaction volumes.

Question 5:Agriculture SmartFarmingProblem

A smart farming system collects soil, weather, and crop data from sensors and drones. Farmers require insights toimprovecrop yieldandresourceusage, while data is generated from multiple distributed sources.

1. Industry Context and Requirements

Collect data from soil sensors, weather systems, drones, and crop observations. Handle distributed and continuous data sources.
Analyze current field conditions.
Predict crop and resource requirements.
Provide practical recommendations for irrigation, fertilization, and crop management.

2. Proposed Big Data Architecture

Precision Agriculture Big Data Architecture

Sensors / Drones / Weather	Data Ingestion	Stream + Batch Processing	DataLake	Analytics / ML	Farmer Dashboard / Recommendations
↓	↓	↓	↓	↓	

3. Technologies and Analytics

Component	Suitable technology / approach	Purpose
Data collection	IoT sensor gateways / APIs	Collect field and weather data
Streaming	Apache Kafka Apache	Receive continuous sensor events Analyze
Processing	Spark	large historical and streaming datasets Store sensor, crop, weather and drone
Storage	Distributed data lake	

		datasets
Crop analytics	Statistical / ML models	Identify relationships between conditions and crop outcomes
Yield prediction	Regression / forecasting models Rule-	Estimate expected crop yield Improve
Resource optimization	based or ML recommendations Farmer	water and fertilizer usage Present
Visualization	dashboard	field insights and recommendations

4. Working

29.Sensors, drones, and weather sources collect field information.
30.The ingestion layer transfers data to the Big Data platform.
31.Streaming processing provides near real-time field information. 32.Historical data is processed to identify long-term patterns.
33.Analytics models estimate crop condition and yield.
34.The system identifies irrigation, fertilizer, or other resource requirements.
35.Recommendationsare displayed to farmers through a dashboard.

5. Security and Risks

Protect farm and operational data using authentication and access controls. Validate sensor data to detect faulty or abnormal readings.
Use backups and fault-tolerant storage for important datasets.
Monitor distributed devices and data sources for failures.

6. Expected Outcome

Thesolutionconverts distributed sensor, weather, crop, and drone data into actionable insights that can help farmers improve yield and use water, fertilizer, and other resources more efficiently.